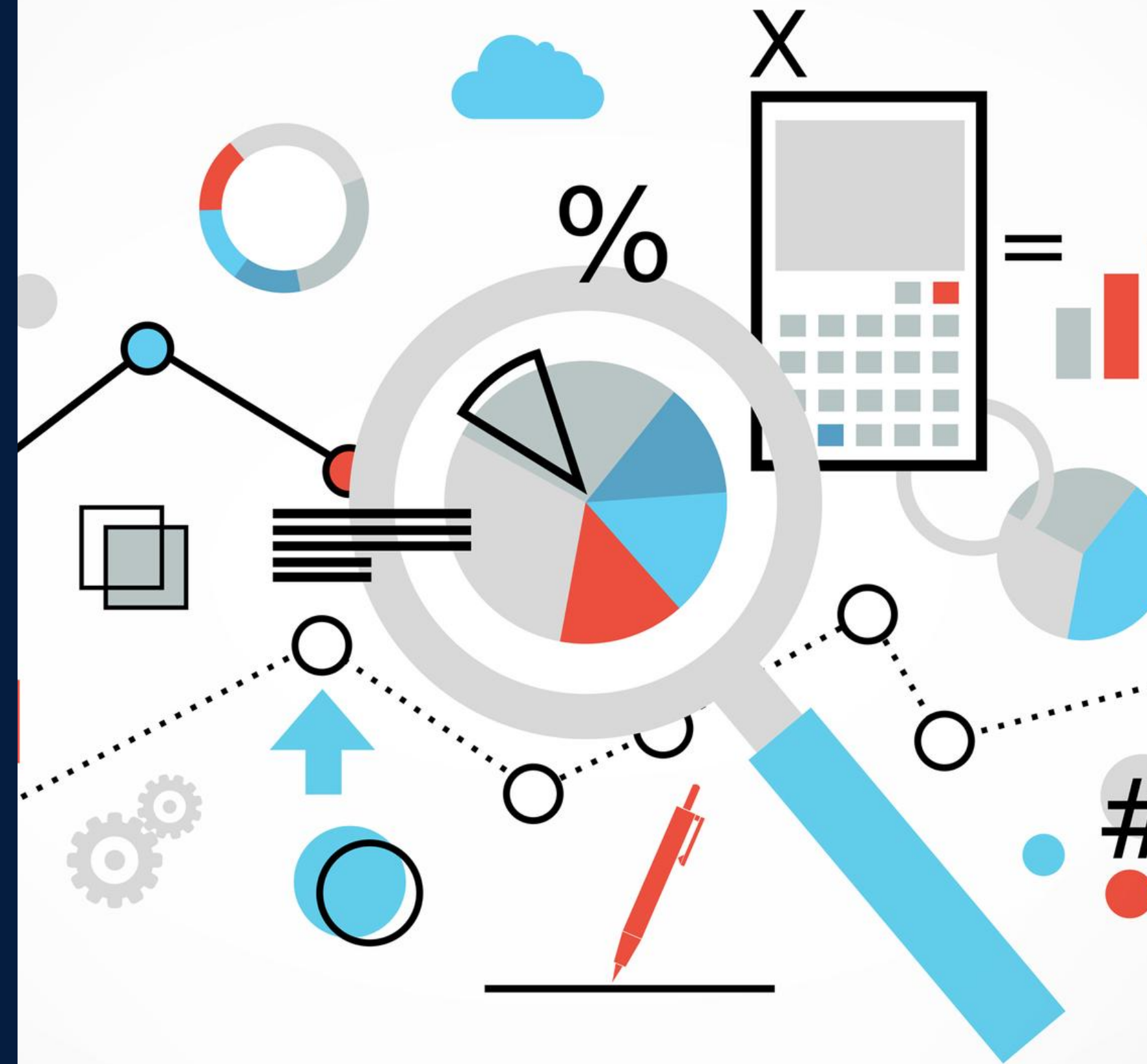




Team:

DATA THIS DATA THAT

*Illinois Statistics Datathon
2025*



Meet Our Team



Ansh

MS In Information
Management



Mohi

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Management



Ishan

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Management



Our Solution to Synchrony Financial's Problem Statement

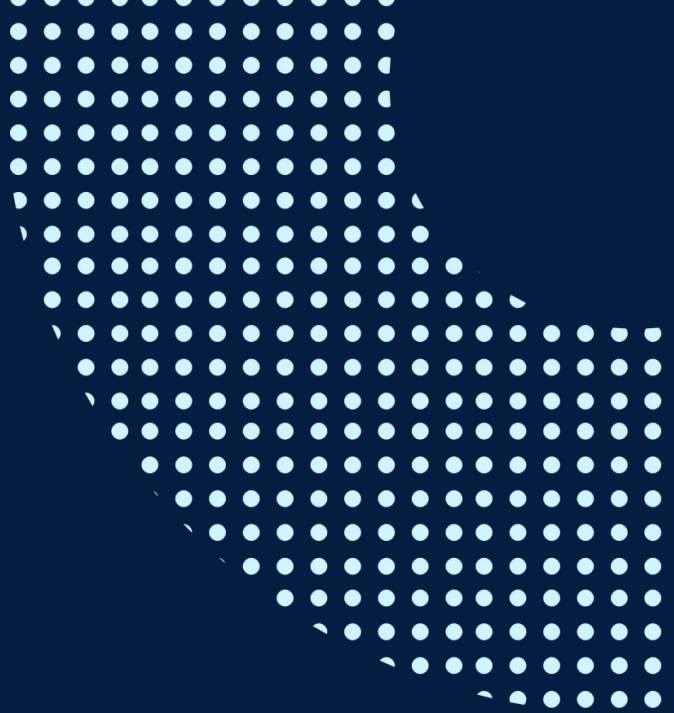
Plan And Methodology

Data Cleaning

EDA

Feature Engineering

Model Training



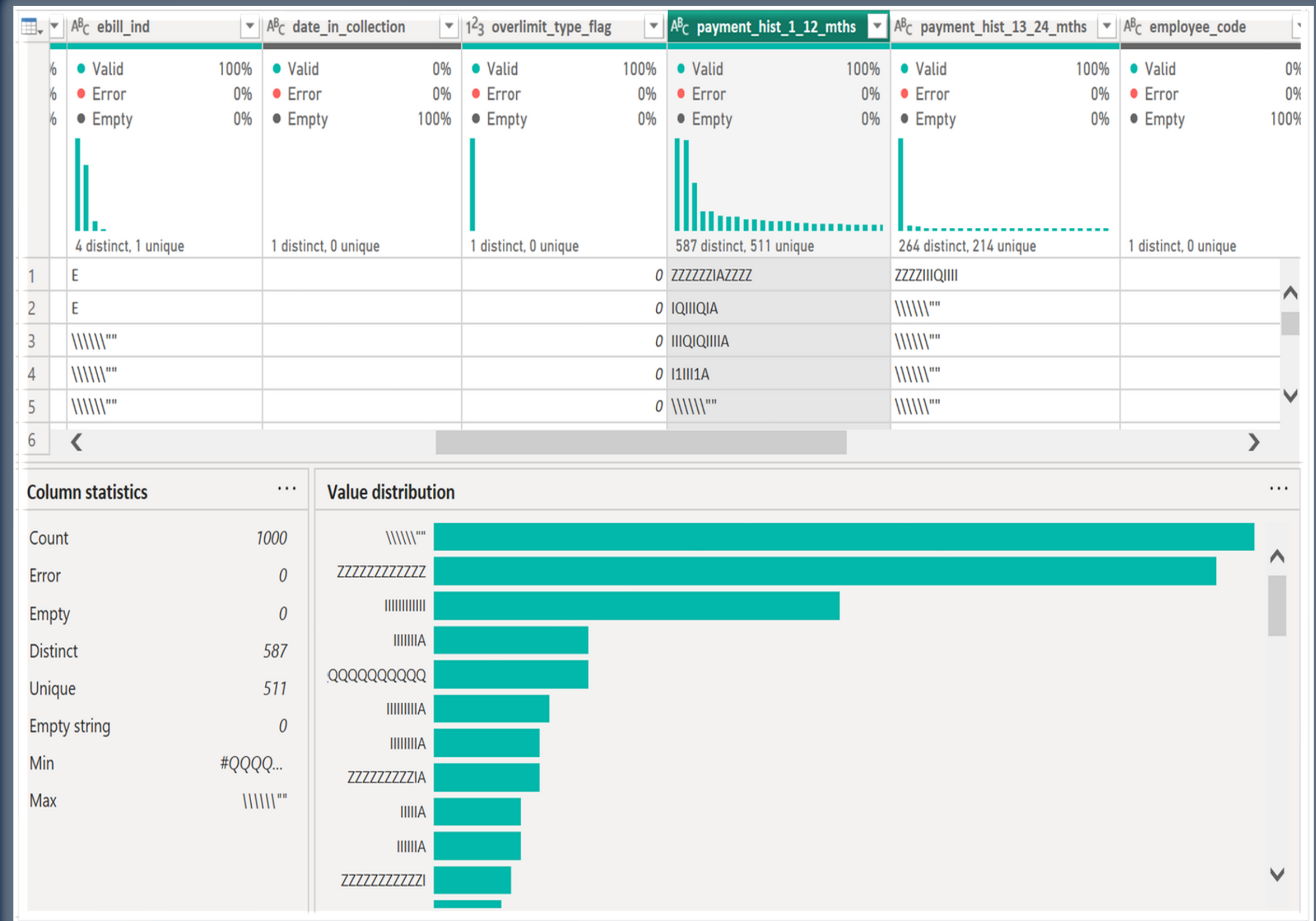
01

Data Cleaning

Data Profiling on Power BI

We loaded all the data to Power BI and used its transform data feature to analyze the following:

- Value distribution
- Column Quality
- Column Statistics
- Column Profile



Cleaning Steps

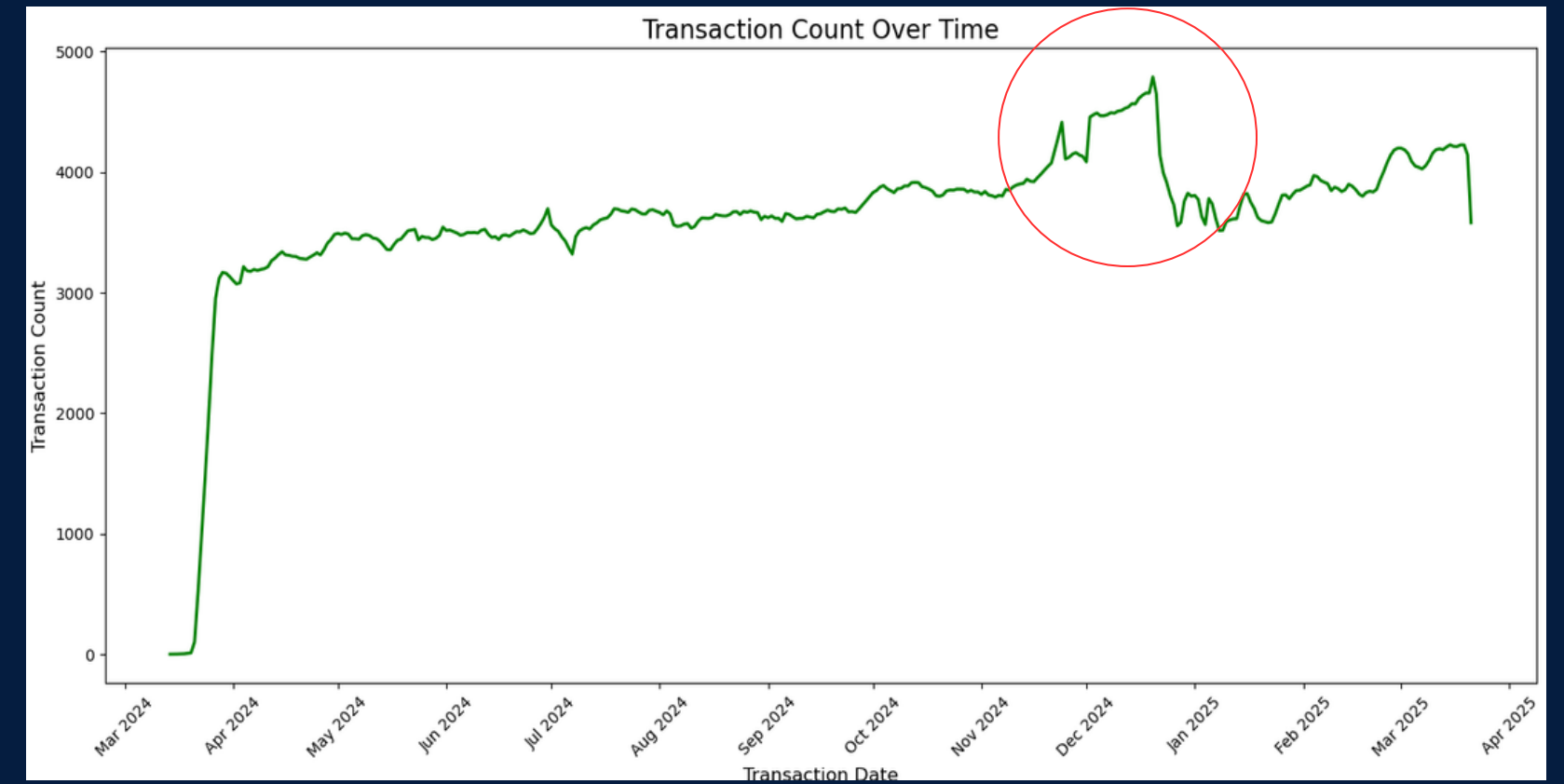
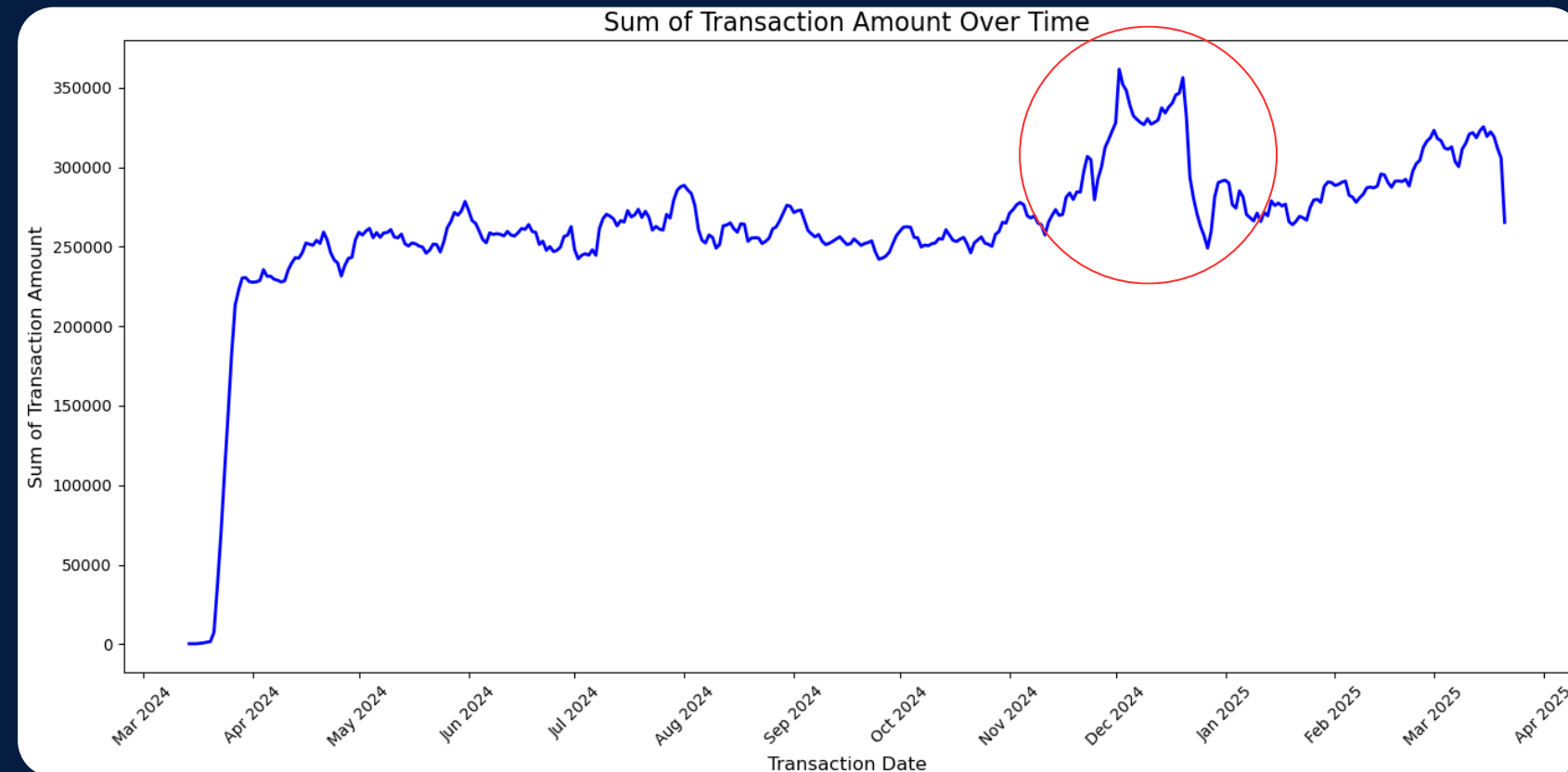
- Removed Redundant and unused columns like:
 - `date_in_collection`
 - `special_finance_charge_ind`
 - `card_activation_date`
 - etc.
- Replaced values such as empty, `999*`, and `///"///` values with `NaN`
- Replaced missing and 0 values for important columns with meaningful values:
Ex: For `Credit Score` we replaced 0 values with the median of the column
- Filtered the datasets for activated accounts only to build model on reasonable data



02

Exploratory Data Analysis

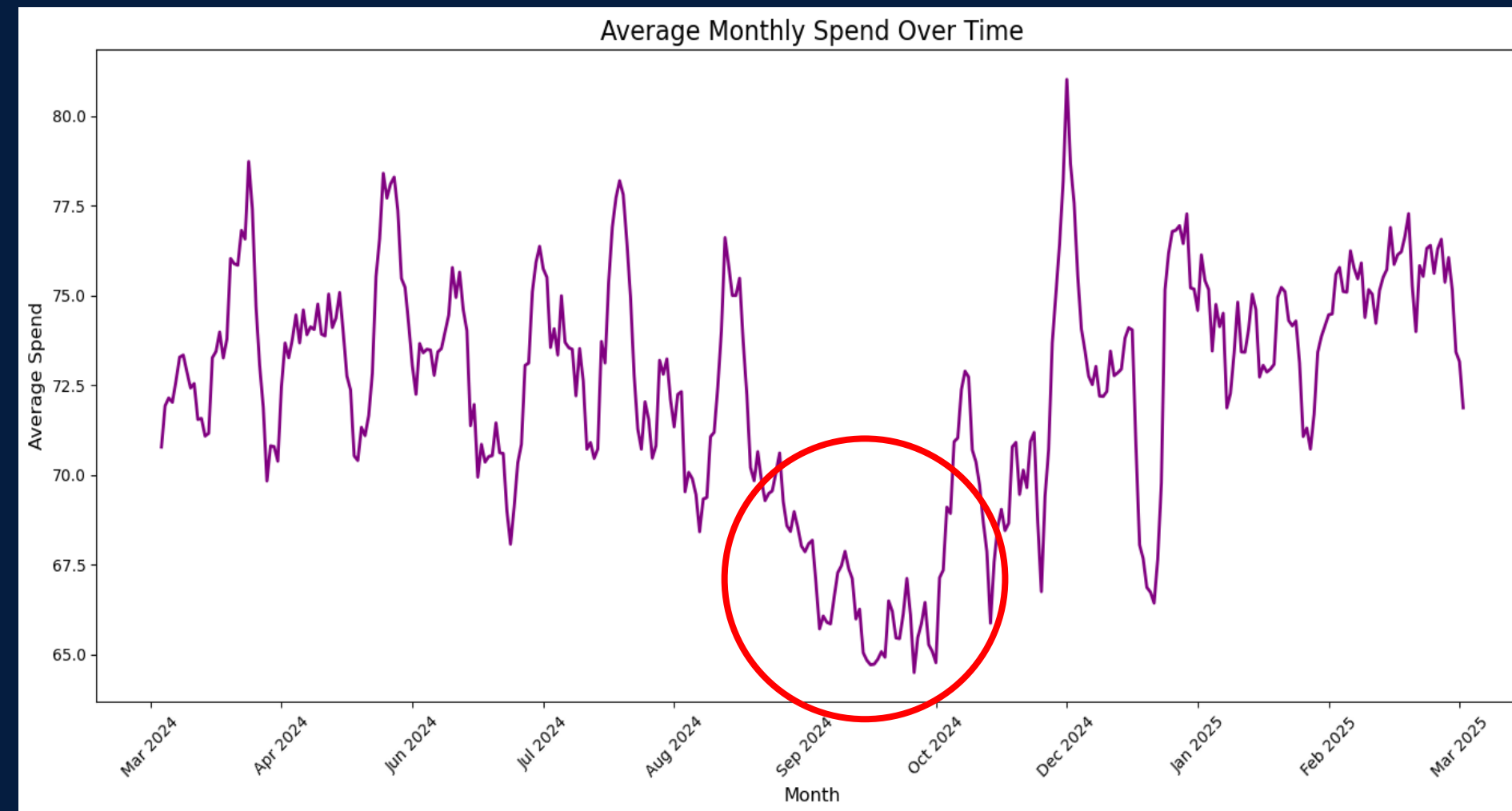
WHY DID TRANSACTION AMOUNTS SPIKE AND DROP OVER TIME?



An increase in total transaction amount and transaction count was observed during the holiday season i.e., Dec to Jan



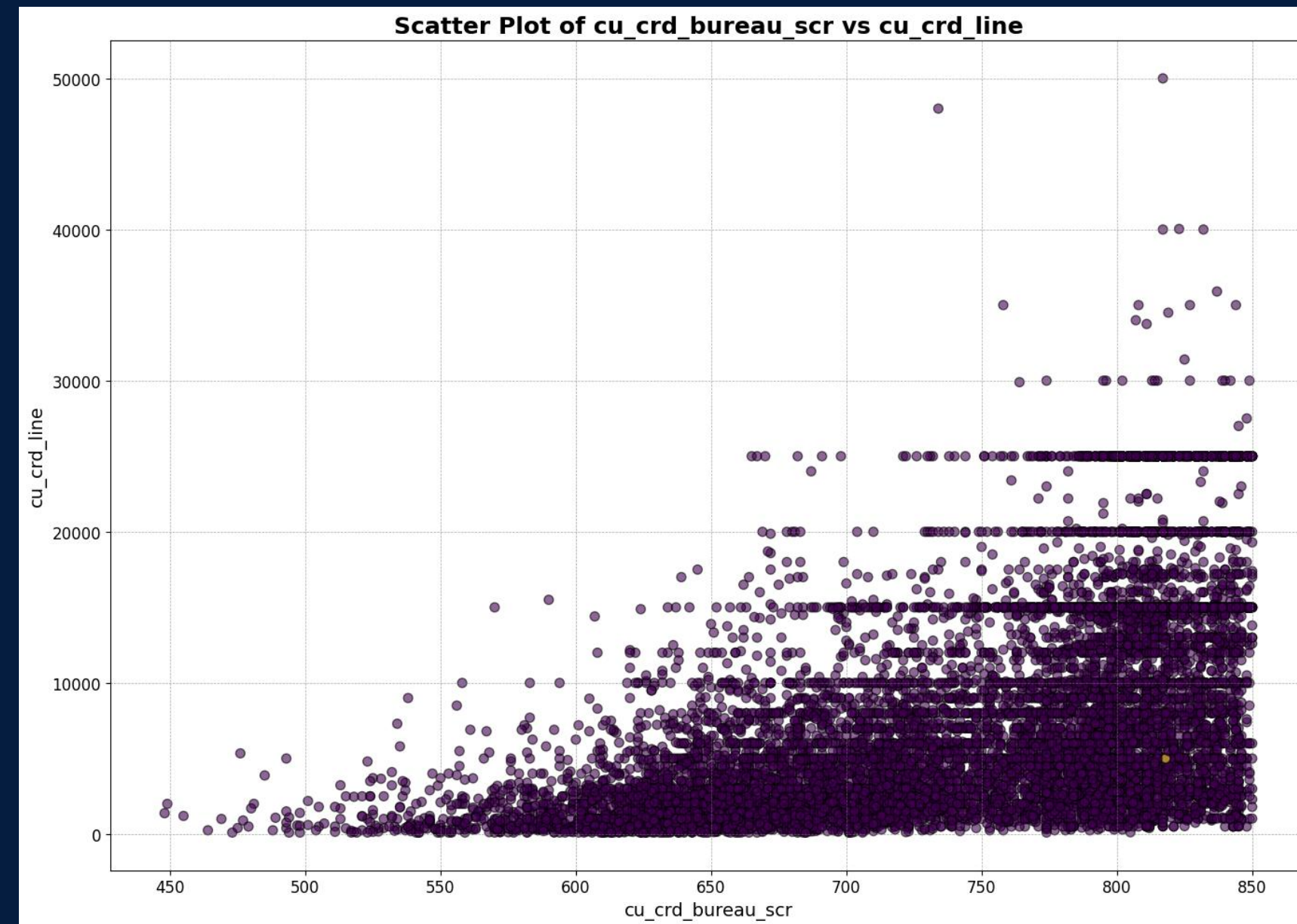
HOW CAN WE EXPLAIN THE FLUCTUATIONS IN AVERAGE MONTHLY SPEND?



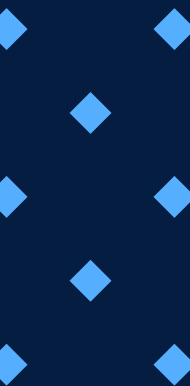
An unusual drop in average monthly spendings was observed during the months of Sept and Oct.

Possibly saving up for holiday season? 😊

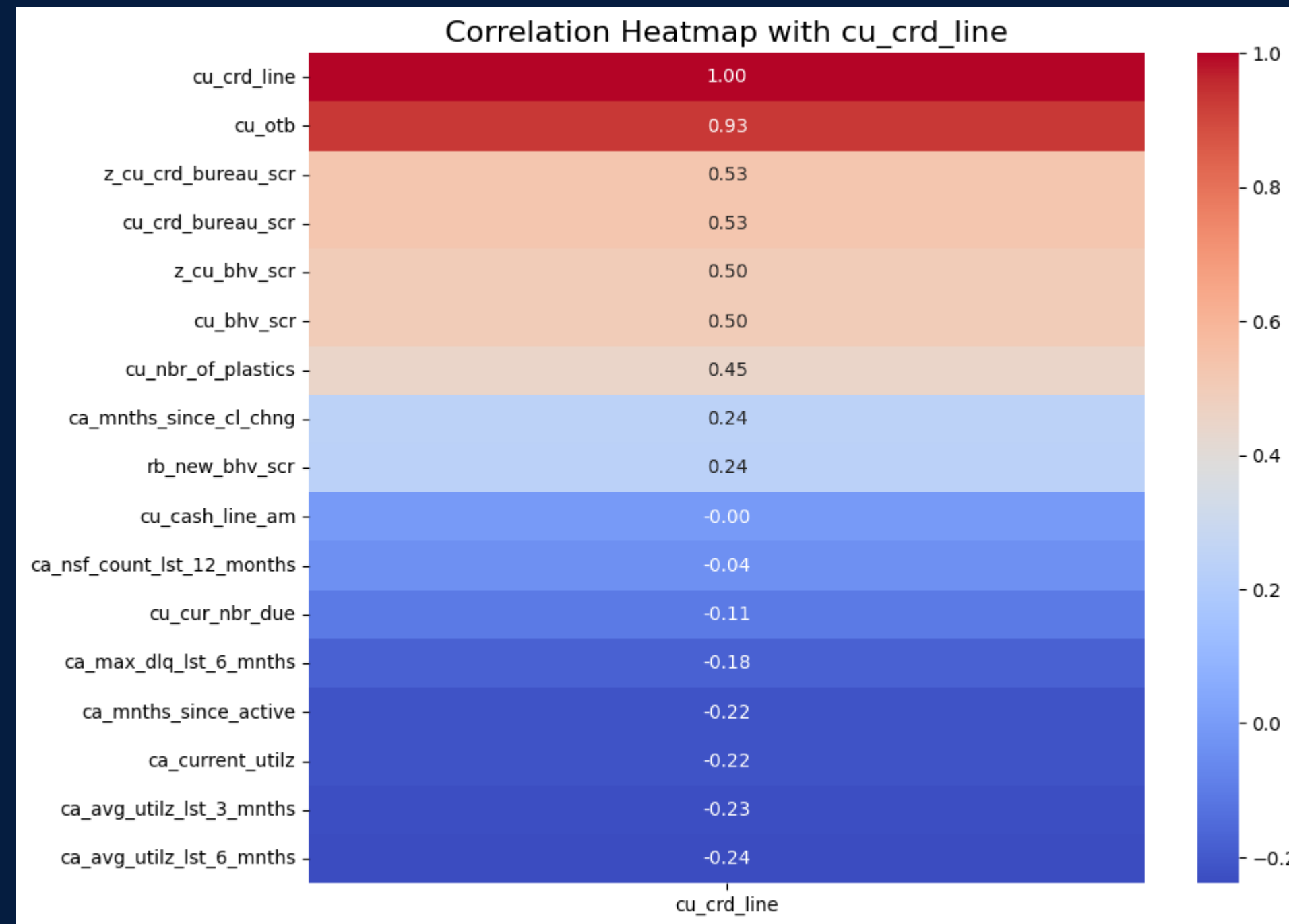
SOME PRE-EXISTING BELIEFS DO HOLD TRUE



Following the popular belief, we could clearly see that the credit score was highly correlated to credit line.



CORRELATION HEATMAP WITH EXISTING CREDIT LINE



Creating a correlation heatmap gave a pretty good idea that features like credit score, behavior score, utilization ratios are great predictors for predicting credit line

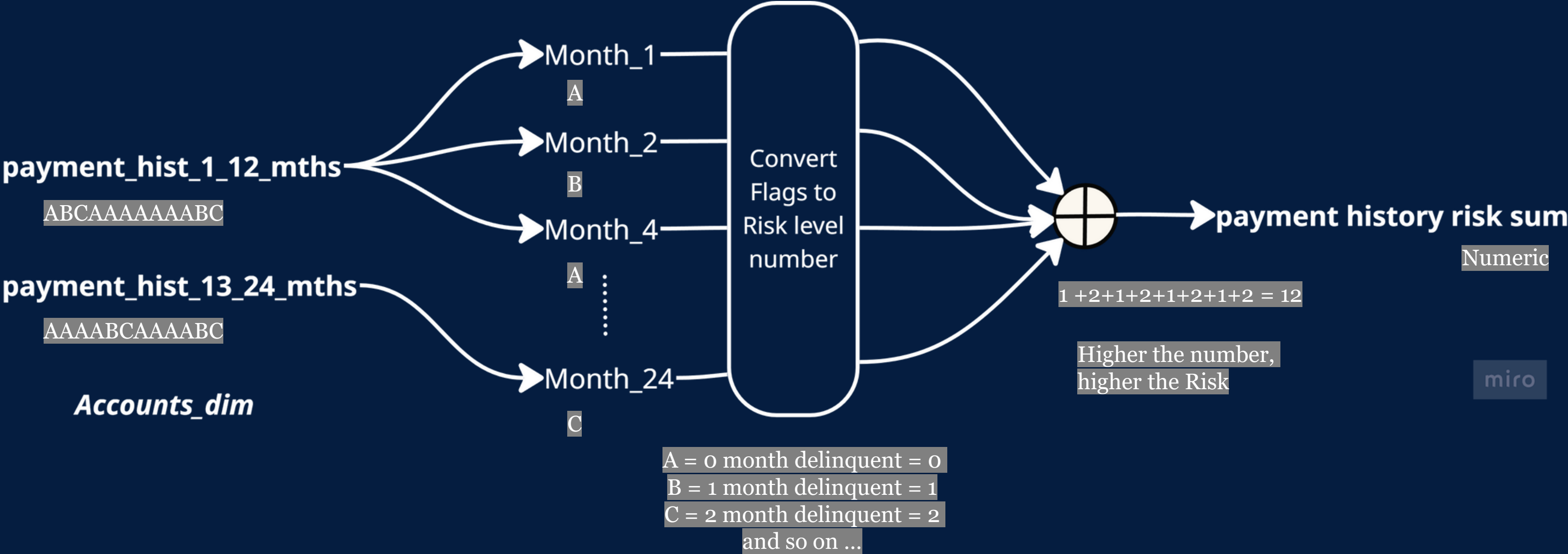


03

Feature Engineering

PAYMENT HISTORY DECODING

Example



We decoded the payment history column and converted them into a single numerical score which could quite significantly predict an accounts risk.

FRAUD DATA UTILIZATION



We also took the fraud dataset and the account which had a fraud case in the past was given a Boolean flag of 1, indicating higher risk. Similar Boolean flag was created if the Fraud case was greater than 1000

MERGED TRANSACTIONS DATA



Since we had 2 different dataset for transaction, we merged them together, joining them on the basis on account no. After that we created new column which gave us the descriptive statistics about the transaction in each account.

04 Model Training



OBJECTIVE 1:

**BUILD A PREDICTIVE MODEL TO FORECAST A CUSTOMER'S
SPENDING FOR Q4 BASED ON PAST YEARS' SPENDING TRENDS
AND RECENT LAST 8 MONTHS OF THE CURRENT YEAR**

SPEND ANALYSIS FORECAST

MODEL:

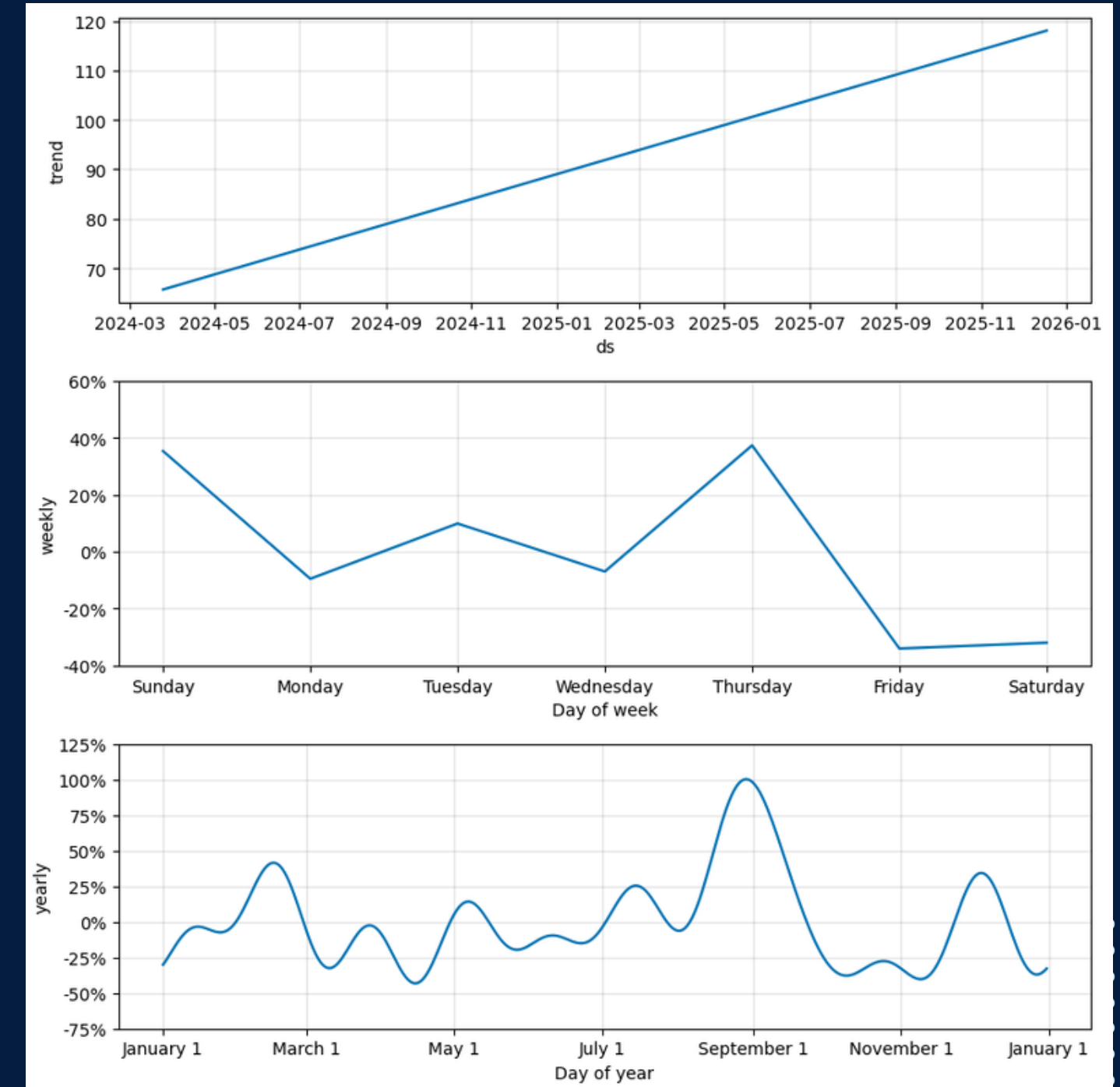
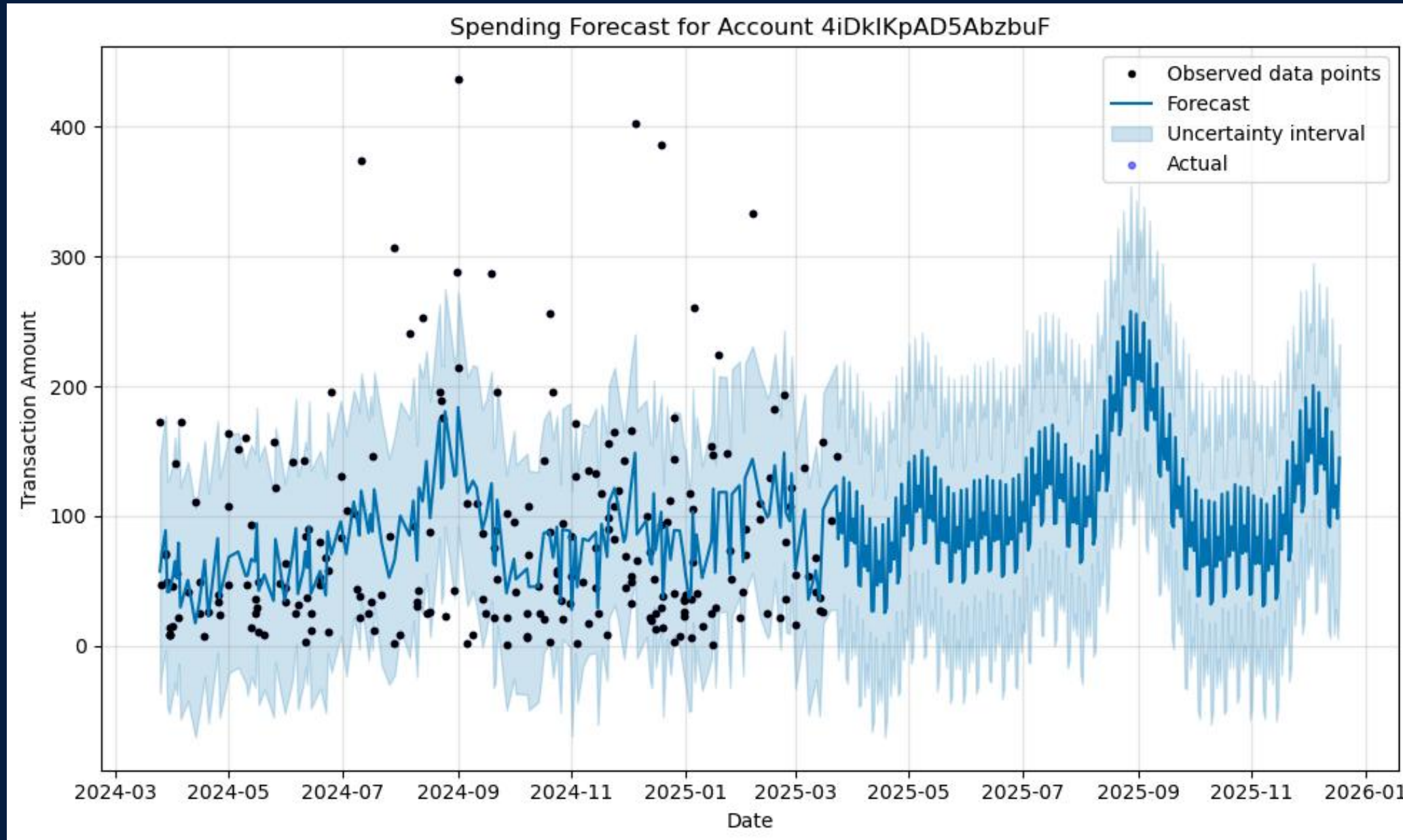
PROPHET

DATASETS USED:

- TRANSACTIONS
- WORLD_TRANSACTIONS

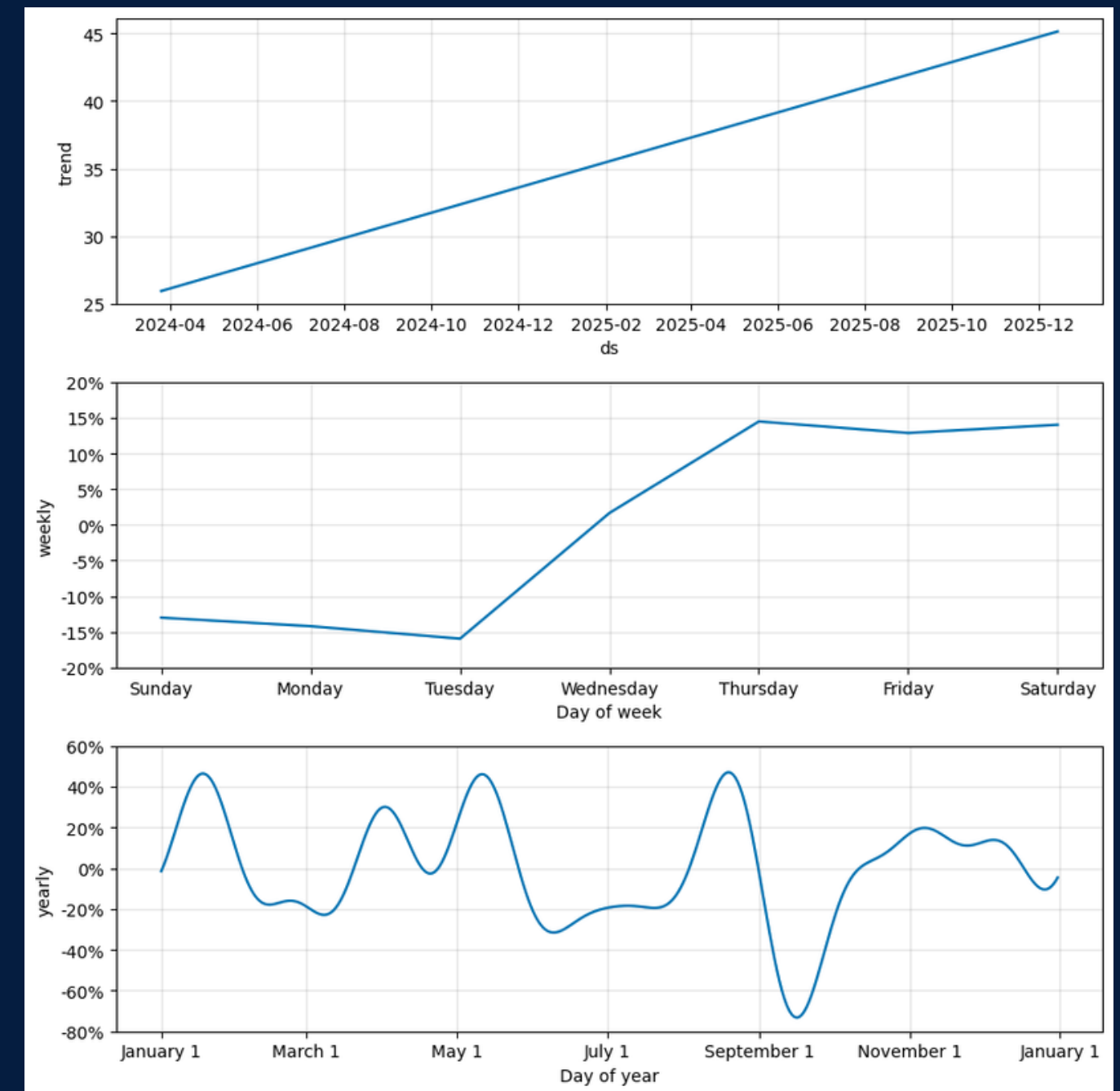
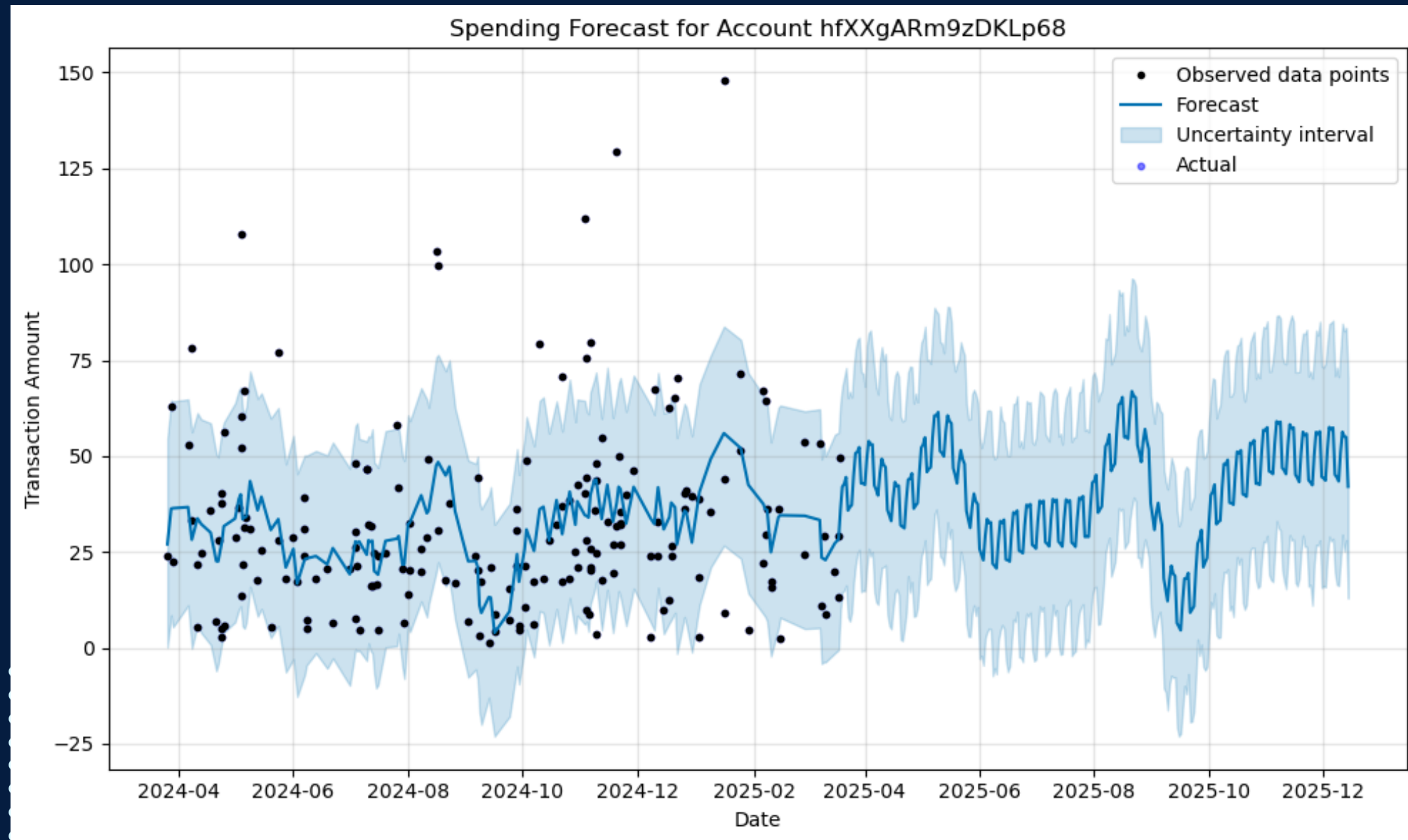
For this problem statement, we chose to follow a time-series forecasting approach and used the Prophet model for easy tunability. The prophet model was run in a loop and was made to fit each individual account's transactions (a really time-consuming step).

Spending Forecast for sample_acc_1



For this sample account we observed a spike during the month of September and this behavior was accurately predicted by the prophet model, showing a similar spike for September 2025

Spending Forecast for sample_acc_2



Contradictory to the previous account, this account showed a dip in the months of September. Since we trained the prophet model on each account, it was able to predict this drip accurately.

OBJECTIVE 2:

**BUILD A CLASSIFICATION MODEL TO SEGMENT THE
CUSTOMERS BASED ON THE ACCOUNT DATA**

CLASSIFICATION MODEL CUSTOMER SEGMENTATION

MODEL:

RANDOM FOREST

DATASETS:

- ACCOUNTS
- FRAUD
- ACCOUNT_FEATURES
- STATEMENTS

From the EDA we have already deduced that some factors like credit score, utilization ratio and behavior score play an important role in deciding the credit line, so we chose to turn the problem into a supervised problem by labelling the data based on the above-mentioned column using knowledge from external sources and feature trends. Random forest was select as the model so that we can easily get the feature importance.

LABEL CREATION

```
classification_model_custor

# Define target labels for classification
def classify_accounts(row):
    if (
        row['ca_avg_utilz_lst_6_mnths'] < 30 and
        row['cu_cur_nbr_due'] == 0 and
        row['return_check_cnt_2yr'] == 0 and
        row['fraud_last_12_months'] == 0 and
        row['cu_crd_bureau_scr'] ≥ 740 and
        row['ca_max_dlq_lst_6_mnths'] == 0 and
        row['ca_mnths_since_active'] ≥ 6 and
        row['ca_mnths_since_cl_chng'] ≥ 6 and
        row['ca_nsf_count_lst_12_months'] == 0
    ):
        return "Eligible for Credit Line Increase without Risk"

    elif (
        0 ≤ row['ca_avg_utilz_lst_6_mnths'] ≤ 60 and
        0 ≤ row['cu_cur_nbr_due'] ≤ 1 and
        0 ≤ row['return_check_cnt_2yr'] ≤ 1 and
        0 ≤ row['fraud_last_12_months'] ≤ 1 and
        600 ≤ row['cu_crd_bureau_scr'] < 740 and
        0 ≤ row['ca_max_dlq_lst_6_mnths'] ≤ 1 and
        row['ca_mnths_since_active'] > 6 and
        row['ca_mnths_since_cl_chng'] > 6 and
        0 ≤ row['ca_nsf_count_lst_12_months'] ≤ 1 and
        row['fraud_amount_gt_1000'] == 0
    ):
        return "Eligible for Credit Line Increase with Risk"

    elif (
        row['ca_avg_utilz_lst_6_mnths'] > 60 or
        row['cu_cur_nbr_due'] > 1 or
        row['return_check_cnt_2yr'] > 1 or
        row['fraud_last_12_months'] > 1 or
        row['cu_crd_bureau_scr'] < 600 or
        row['ca_max_dlq_lst_6_mnths'] > 1 or
        row['ca_nsf_count_lst_12_months'] > 1 or
        row['fraud_amount_gt_1000'] > 0
    ):
        return "Non-Performing Account"

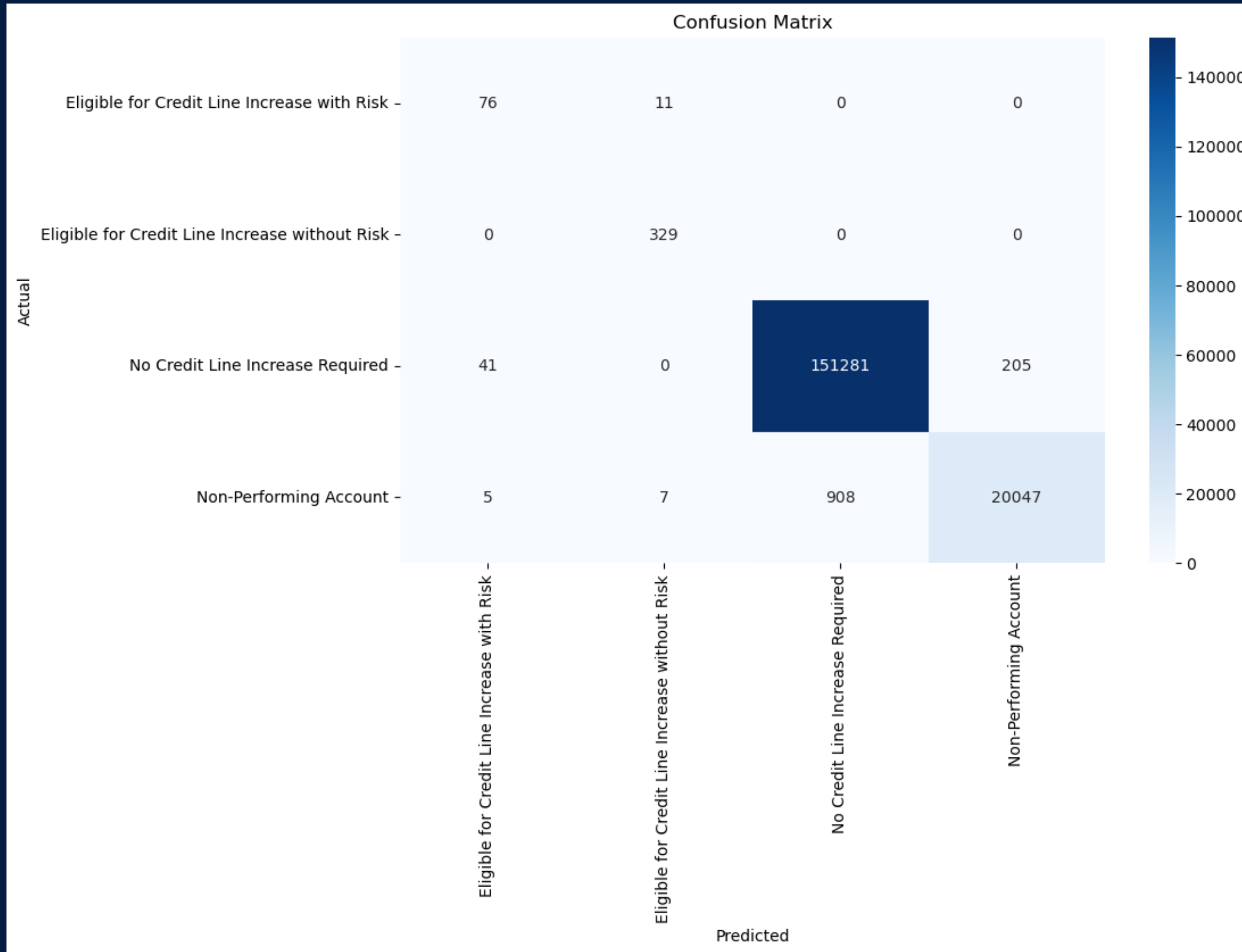
    else:
        return "No Credit Line Increase Required"
```

We chose a subset of features and decided to manually apply certain condition to classify the accounts into the following categories:

- Eligible for Credit Line Increase without Risk
- Eligible for Credit Line Increase with Risk
- Non-Performing Account
- No Credit Line Increase Required

These labels were then used to train a Random Forest Classifier.

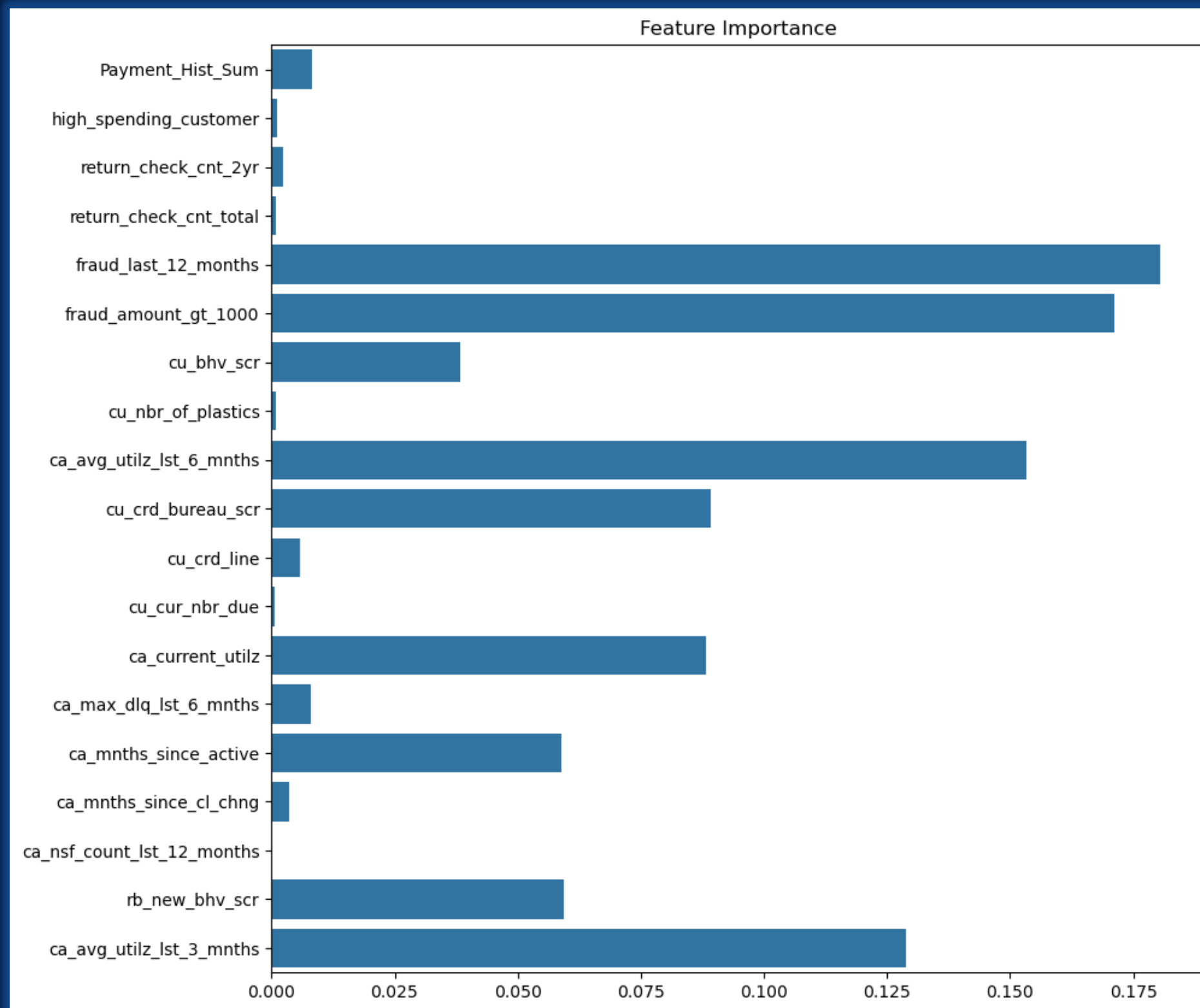
Model's Confusion Matrix



Since the features we used to pre-label the data was already present in the dataset, we knew that classification accuracy of the model would be high, which can be seen in this confusion matrix.

What we cared about was which other features did the model prioritize to classify the data.

Feature Importance



We found out that payment history was also important in determining the credit line increase risk of an account.

OBJECTIVE 3:

BUILD A MODEL THAT HELP IDENTIFY CREDIT LIMIT
ADJUSTMENTS BASED ON SPENDING PATTERNS
AND RISK FACTORS.

CLASSIFICATION MODEL CUSTOMER SEGMENTATION

MODEL:

GRADIENT BOOSTED REGRESSOR

DATASETS:

- ACCOUNTS
- FRAUD
- ACCOUNT FEATURES
- STATEMENTS
- TRANSACTION
- WORLD_TRANSACTION

For this objective as well, we decided to go with the gradient boosted tree and consider this problem as a supervised problem. This model ended up using data from almost all the datasets available to us.

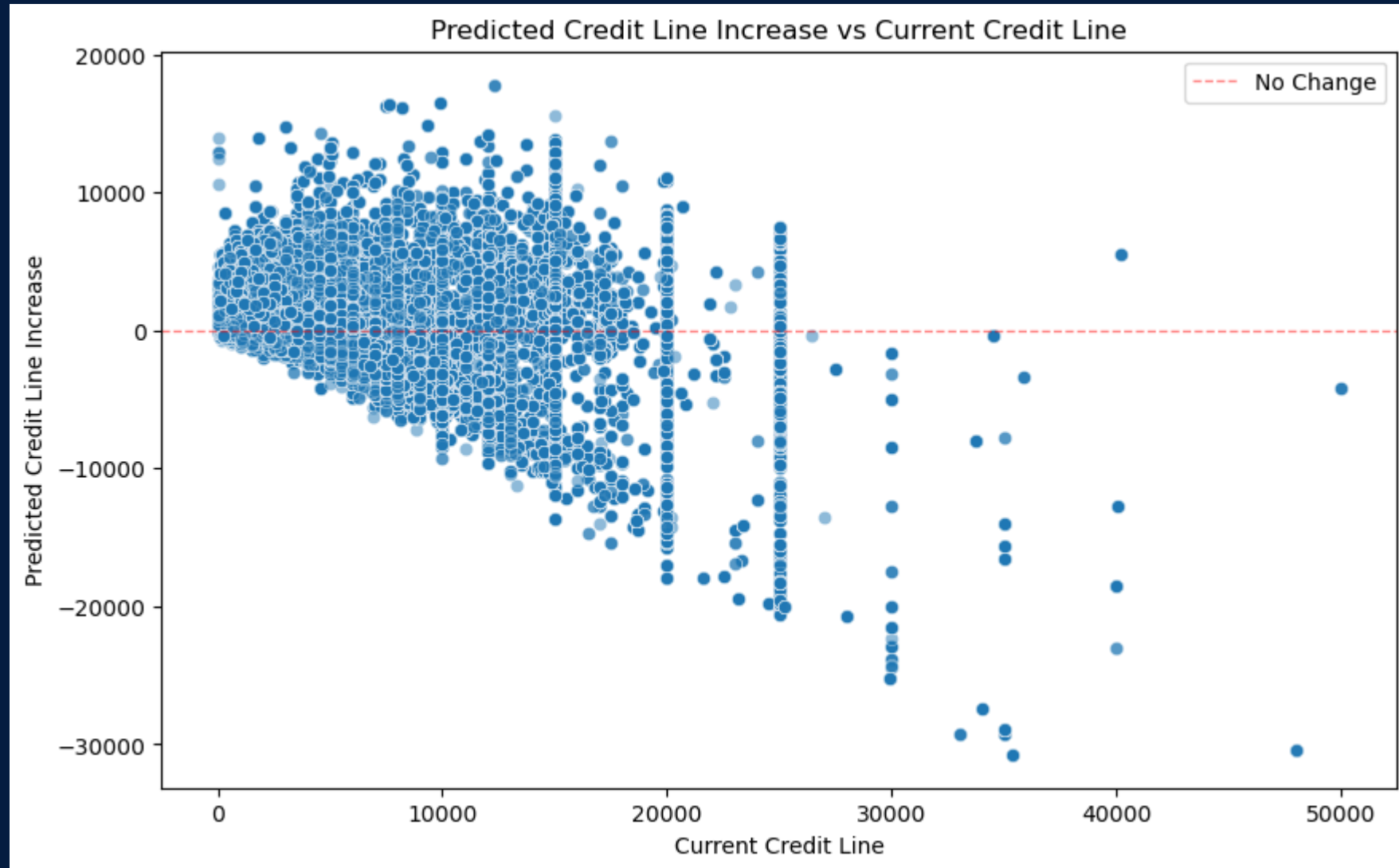
LABEL CREATION

```
credit_line_adjustment_multipl

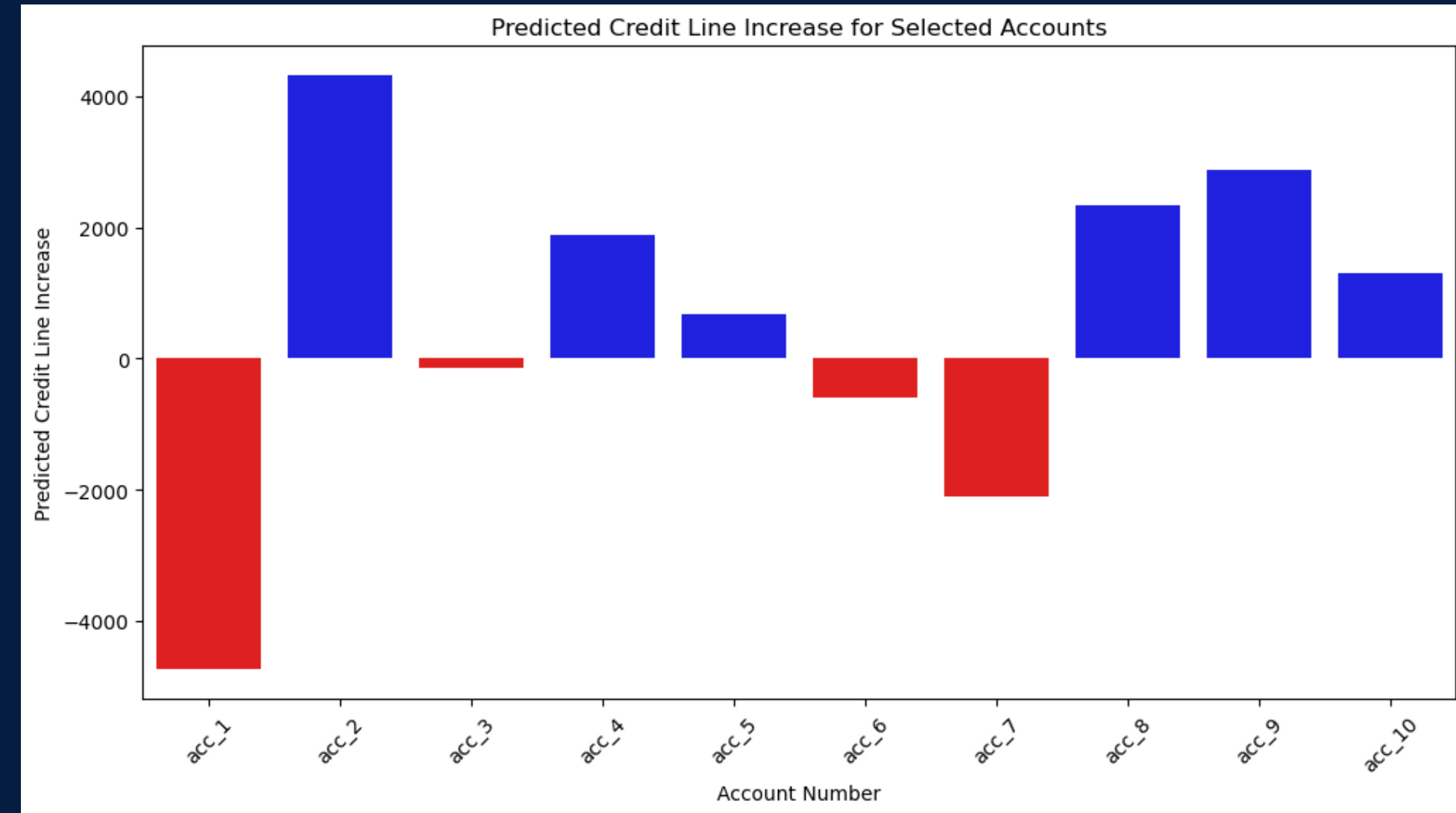
# Define target variable (recommended credit limit adjustment)
def calculate_credit_limit_adjustment(row):
    # Example formula: Adjust based on utilization, delinquency, and fraud indicators
    if row['fraud_amount_gt_1000'] > 0 or row['fraud_last_12_months'] > 1:
        return row['cu_crd_line'] * 0.8 # Reduce credit limit by 20%
    elif row['ca_avg_utilz_lst_6_mnths'] > 50 or row['cu_cur_nbr_due'] > 2:
        return row['cu_crd_line'] * 1.2 # Increase credit limit by 10%
    elif row['transaction_amt_max'] > 0.5 * row['cu_crd_line'] and row['ca_max_dltq_lst_6_mnths'] < 50:
        return row['cu_crd_line'] * 1.4
    else:
        return row['cu_crd_line'] * 1.1 # Increase credit limit by 20%
```

We chose a subset of features and based on that we decided the multiplier on the credit line. Using this we created new field called recommended credit limit adjustment which could be used as label to train our Gradient Boosted Regressor.

MODEL'S OUTPUT

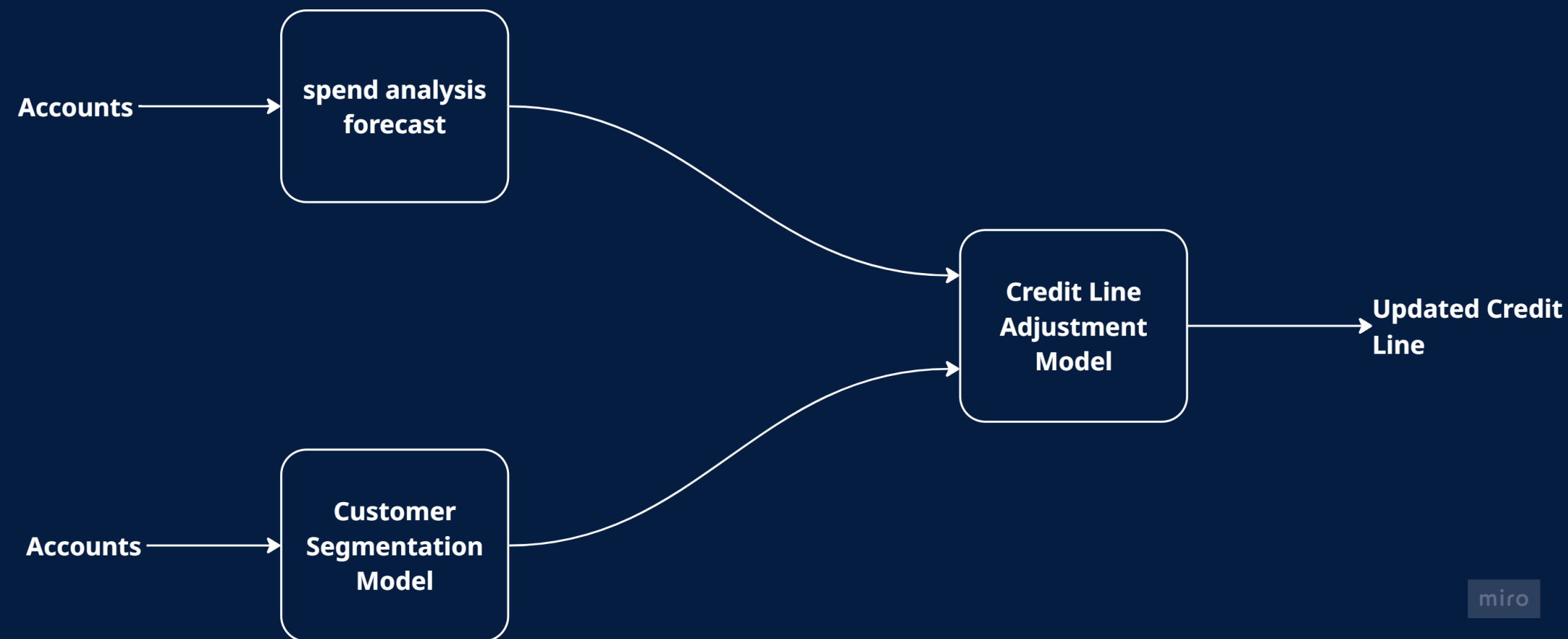


Model was able to generalize well and generate by how much the credit line of an account should be increased based on all the predictors from the datasets.



We could see that the model not only suggested credit increase but also suggested that the credit line of some accounts should be decreased.

ENHANCED FRAMEWORK PROPOSAL



We even had a plan to plug these models together to build a meta model which do all the three task in one go and provide an even better credit line change recommendations. But due to the time constraints of the Datathon we were not able tot achieve this goal.

THANK
YOU